Title: The Curious Case of Aarav

Aarav was a curious computer science student living in Lucknow. He spent most of his days experimenting with code, trying to build small applications that could solve real-world problems. One day, he decided to create a PDF summarizer using artificial intelligence.

He started by learning how text is extracted from PDF files using tools like PyMuPDF. At first, the extracted text looked messy and unstructured, but Aarav realized that breaking the text into smaller chunks could help. This led him to explore the concept of chunking, where large text is divided into smaller pieces for easier processing.

Next, Aarav discovered embeddings. He learned that embeddings convert text into numerical vectors, which can be stored in a vector database like FAISS. This allowed him to perform semantic search, meaning the system could find relevant information based on meaning rather than exact keywords.

To generate answers, Aarav used a transformer-based language model. Initially, the model gave repetitive and unstructured responses. However, by refining prompts and controlling parameters like do_sample=False, he improved the quality of responses significantly.

For summarization, Aarav first tried simply taking the first few lines of the document, but he realized this was not true summarization. Later, he implemented a proper summarization model using Hugging Face Transformers, which generated concise and meaningful summaries.

After several days of debugging, testing, and improving the user interface, Aarav finally completed his project. The application allowed users to upload a PDF, view a summary, and ask questions about the document.

In the end, Aarav learned that building AI systems is not just about using models, but about combining multiple components—text extraction, chunking, embeddings, vector search, and language models—into a cohesive system.